

Procurement is an engineering problem.

Give it an engineer.

A production line is 12 days from standing still. Stockout reads the ERP, finds suppliers, dispatches voice agents, runs a landed-cost model, checks compliance, and ships the decision as a pull request.

[Open the cockpit →](#)[View a sample PR](#) sourcing_case.yaml

```
shortage:
  part: 6204-2RS
  qty_needed: 5000
  needed_by: 2026-09-02
```

```
candidates:
```

A shortage costs three things

It is slow

Days of phone tag per part. **€18,400 an hour** if the line stops, and the line does not wait for the phone to be answered.

The decision is worse than it looks

A buyer calls the two or three vendors they already know. Nobody weighs fifteen suppliers against MOQ, Incoterms, freight, duty and carrying cost by hand.

So it is more expensive

Least of all does anybody hand-compute a **split order** — and that is usually where the money is.

The option nobody has time to find

```
candidates:
- skf_germany:      compliance: PASS
- fag_italy:        compliance: PASS
- ntn_china:        compliance: FAIL (blocked_origin)
- generic_turkey:   compliance: FAIL (missing_cert)

recommended:
split:
- air 20% from SKF   # covers the line stop
- sea 80% from FAG   # unit economy
```

Air the fifth that saves the line. Ship the rest.

One agent works the whole case

Reads	the system of record — bin, take rate, the BOM line that stops, the incumbent, the slipped PO
Screens	every supplier against the company's own compliance rules, rejected by name and by the rule that failed
Calls	them through CALL-E — MOQ, Incoterms, lead time, price breaks
Costs	every single-source and split plan, landed, on-time-first-then-cheapest
Ships	the decision as a pull request

Watch it work

stockout

CASE-001

6204-2RS

LINE STOPS IN

DETECTED

PARTS AT RISK

5 below reorder point · 2 with an open case

DIN933-M8x40-A2

9.8d cover

Hex head bolt M8x40 A2-70

22,000 on hand / 45,000 reorder

€9,200/h if the line stops

6204-2RS

12d cover

Deep-groove ball bearing 6204-2RS

4,200 on hand / 9,000 reorder

€18,400/h if the line stops

6205-2RS

10.9d cover · no case

SIM-25x52x7

19.6d cover · no case

DIN912-M8x30-A2

25.6d cover · no case

Shortage

Suppliers

Candidates

Calls

Cost

Decision

DETECTED

RESEARCHING

CALLING

01 THE SHORTAGE

Requires Attention

Launch sourcing agent

NEEDED

36,000

by 03 Sept 2026

ON HAND

4,200

31,800 short

LINE STOPS IN

.

03 Sept 2026

COST OF STANDING

STILL

€18,400/h

PLANT-MUC · ASSY-3

6204-2RS

Deep-groove ball bearing 6204-2RS

rolling_bearing

criticality: high

Single-row deep-groove ball bearing, 20×47×14 mm, rubber-sealed both sides, DIN 625-1.

STANDARD	BORE MM	OUTER DIAMETER MM	WIDTH MM
DIN 625-1	20	47	14
SEAL	CLEARANCE	DYNAMIC LOAD RATING	MAX SPEED RPM
2RS (contact rubber, both sides)	C3	KN 12.7	14000
WEIGHT	HS CODE		
0.102 kg	8482.10.10		

Reason: PO-2291 (30 000 pcs, Kugellager Bayern) slipped from 2026-08-26 to 2026-09-17. On-hand cover is 12 days at the ASSY-3 take rate of 350/day.

WHAT THE AGENT IS DOING

17/20 done

Reading the incident

✓ Reading the shortage the ERP raised

PO-2291 (30 000 pcs, Kugellager Bayern) slipped from 2026-08-26 to 2026-09-17. On-hand cover is 12 days at the ASSY-3 take rate of 350/day.

Pulling part data

✓ Pulling the part record, spec, weight and HS code

6204-2RS, 0.102 kg, HS 8482.10.10

✓ Reading on-hand stock, reservations and take rate

4,200 on hand at PLANT-MUC

✓ Checking open purchase orders for a delivery

PO-2291 (30 000 pcs, Kugellager Bayern) slipped from 2026-08-26 to 2026-09-17. On-hand cover is 12 days at the ASSY-3 take rate of 350/day.

✓ Reading what we paid for it before

what we paid before, by quarter

Finding registered suppliers

...and the phone in the room is the same agent, live.

Where this goes

One AI for the whole supply chain.

End to end, inside the ERP the buyers already live in.

Shortage firefighting is the first workflow — the one that earns the trust.

Then reorder points, supplier discovery, price negotiation, risk on the whole bill of materials.

How we get there

Start narrow, and in person

A handful of Munich manufacturers. One part family at a time, in the plant, with their own compliance rules.

Then compound

Every case adds suppliers, quotes, call playbooks and rules that the next customer inherits. The tenth plant is cheaper to serve than the first.

Being local is the strategy, not a limitation — we can stand
on the shop floor the day the line is at risk.

Nothing is ordered.
A human merges the PR.

stockout · EHL Game Jam Munich · Cognition "Devin for X"

Backup slides — question round

Guardrails

- **Nothing is ordered.** The agent ranks and stops; a buyer picks.
- **Unknown stays unknown.** Every field a call did not establish reads `unknown` — it is never inferred.
- **Too late is too late.** A plan that misses the line stop is shown as too late however cheap it is.
- **Approve by merge.** The pull request is the human-in-the-loop gate.
- **The call discloses itself.** Every supplier is told they are speaking to an AI, before anything is asked.

Under the hood

- Contracts first: one frozen set of models shared by every slice.
- The system of record sits behind one interface — mock YAML today, SQL today, a real ERP adapter tomorrow, same tool URLs.
- The cost model is pure functions with a passing test suite: landed cost, freight, duty, carrying cost, split orders.
- `cases/` is the datastore — one directory per case, append-only event log. The artifact *is* the record.
- 116 tests, and the cockpit runs offline on fixtures.

What we do not claim

- No purchase is placed. No supplier is committed to anything.
- A live agent session takes tens of minutes; the sub-second run you saw is the deterministic backend, and the recording says which is which.
- The ERP is seeded, not a customer's production system.
- Web search, email RFQ and the automatic shortage detector are designed, not demonstrated.